

PABLO WILLIAMS

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WORK EXPERIENCE

Data Analyst, APJ Kapital

Sep 2023 - Present

London, UK

- Analyse company filings (10-Ks, 10-Qs, 8-Ks), earnings-call transcripts, and real-time market news to produce structured investment signals and buy/hold/sell recommendations for a portfolio of over 30 equities.
- Build and maintain SQL data pipelines in PostgreSQL and Tableau dashboards that track portfolio P&L, sector allocation, and factor exposures for weekly portfolio review meetings.
- Designed an n8n automation workflow that ingests daily market-update videos via Discord, transcribes them with Whisper, summarises with GPT-4o, and distributes briefs to Telegram and email.

AI Investment Programming Intern, MDOTM

May 2022 - Oct 2022

Milan, Italy

- Built AI-driven investment tools and customer-segmentation models in Python and R, applying preprocessing, feature engineering, and supervised learning to sharpen predictive accuracy and targeting.
- Worked with a 40-person cross-functional team to embed ML into investment workflows, running data-quality reviews and turning stakeholder feedback into measurable performance gains.
- Played a key role in MDOTM's ML platform, which won the 2022 AIFIN Financial Innovation Award and was named Best Product of the Year at the New York FinTech Awards.

PROJECTS

[Calibration-Optimised Equity Outperformance](#) | Python, XGBoost, LightGBM, Optuna, SHAP

Jan 2026

- Tested whether ECE-optimised classifiers deliver better portfolio returns under half-Kelly sizing across 457 S&P 500 stocks (2010-2024). Random Forest achieved the best calibration (ECE = 0.026), a Sharpe of 0.77, and 68.1% total return.

Polish Corporate Bankruptcy Prediction | Python, LightGBM, XGBoost, Optuna, SHAP

Dec 2025

- Built a leak-proof ML pipeline on 10,503 company-year records with 64 financial ratios, handling 50% missingness and 20:1 class imbalance. LightGBM reached an AUPRC of 0.6249 (13.3x over baseline) and AUC-ROC of 0.926.

[RiskTerrain](#) - Geospatial S&P 500 Risk Intelligence | Python, LangGraph, SurrealDB, Docker

Mar 2026

- Created a 6-node LangGraph pipeline that ingests live geopolitical events, runs multi-hop supply-chain traversal across 154 S&P 500 companies, and scores cascading risk in under 8 seconds. Built at the LangChain x SurrealDB London Hackathon.

[AlphaQuant Factor Timing](#) | Python, XGBoost, scikit-learn, cvxpy

Sep 2024

- ML-enhanced factor timing model blending Value, Momentum, Quality, and Low Volatility with XGBoost regime detection. Backtested Sharpe of 0.94 (47% above S&P 500) and maximum drawdown cut by 46%.

[Sentinel Credit Risk Prediction](#) | Python, XGBoost, LightGBM, SHAP

Dec 2024

- Ensemble gradient boosting model for consumer loan default prediction on 15,000 Lending Club records. Achieved AUC-ROC of 0.89 with Platt-scaled probabilities and SHAP-based explainability dashboards.

[Vista Zero-Shot Fashion Detection](#) | Python, OWL-ViT, CLIP, YOLOv8

Jan 2024

- Zero-shot detection using CLIP and OWL-ViT with knowledge distillation into YOLOv8-nano. Hit 88.7% of supervised performance (mAP@0.5: 0.751) with 99% annotation cost reduction across 44,441 images.

[Vertex SaaS Churn Prediction](#) | Python, XGBoost, Survival Analysis

Feb 2024

- B2B churn prediction on 20,000 customer records reaching AUC-ROC of 0.91 and precision of 0.85, flagging at-risk customers through engagement metrics and behavioural patterns.

[Project Lockdown](#) - Emergency Food Distribution | Python, VRP Optimisation, Google Maps API

Jul 2024

- Designed an emergency food distribution system for 52,000 residents over 28 days, using exponential smoothing for demand forecasting and vehicle routing optimisation with capacity constraints.

EDUCATION

MSc Business Analytics - UCL

Sep 2025 - Sep 2026

Predicted: Distinction (72%)

Modules: NLP, Machine Learning for Domain Specialists, Predictive Analytics, Data Visualisation, Statistical Methods, Mathematical Analysis.

BSc Business Analytics (Minor: Accounting & Finance) - Lancaster University

Sep 2021 - Jul 2025

Achieved: Upper Second-Class Honours (2:1). Minor: Upper Second.

Modules: Data Mining, Data Visualisation, Spreadsheet Modelling, Statistical Analytics.

SKILLS

Programming: Python (pandas, NumPy, scikit-learn), SQL (PostgreSQL, DuckDB), R (tidyverse, ggplot2), JavaScript

ML: XGBoost, LightGBM, Random Forest, PyTorch, TensorFlow, LLMs (LangChain, Ollama), Optuna, SHAP

Data Engineering: PySpark, MongoDB, DuckDB, Parquet, Docker, n8n, AWS (S3)

Visualisation & BI: Tableau, Streamlit, Matplotlib, Seaborn, LaTeX | **NLP:** Sentence-Transformers, RAG, Whisper, GPT-4o

CERTIFICATIONS

Bloomberg: Finance Fundamentals, Market Concepts, Spreadsheet Analysis | Google: Data Analytics, IT Automation with Python

IBM: BI Analyst, Applied Software Engineering, AI Enterprise Workflow | Yale: Financial Markets | LBS: Financial Analysis | Duke: MLOps

VOLUNTEERING & LEADERSHIP

Mathematics & CS Tutor

Feb 2024 - Present

GCSE, IB, SAT, GMAT levels; tracking student progress with Excel and Tableau diagnostics.

Data Science Society Member

Oct 2022 - May 2025

Assisted peers with ML and analytics projects weekly.

Volunteer, Milan Dog Kennel

Sep 2015 - Sep 2021

Walking and caring for rescue dogs every Sunday.

LANGUAGES

Italian: Native

English: Native

Spanish: Fluent

German: Beginner